



HISTORY OF UNIX / LINUX

DV1457-1578 Programming in UNIX Environments

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BTH

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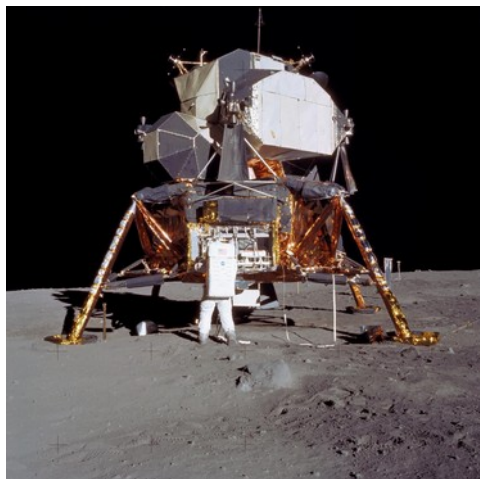
Program

- The birth of UNIX.
- The UNIX Wars.
- The birth Linux.



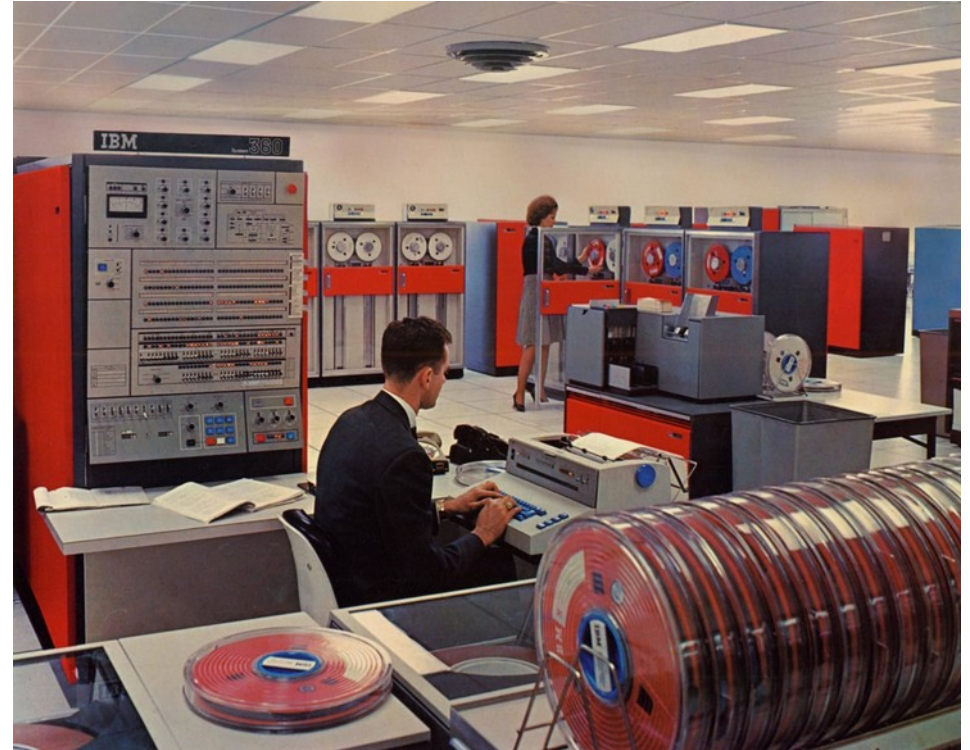
THE BIRTH OF UNIX

Welcome to 1969!



Welcome to 1969!

- Main frames and computers run custom-made operative systems.
- Most popular computer then: IBM System/360.



Welcome to 1969!

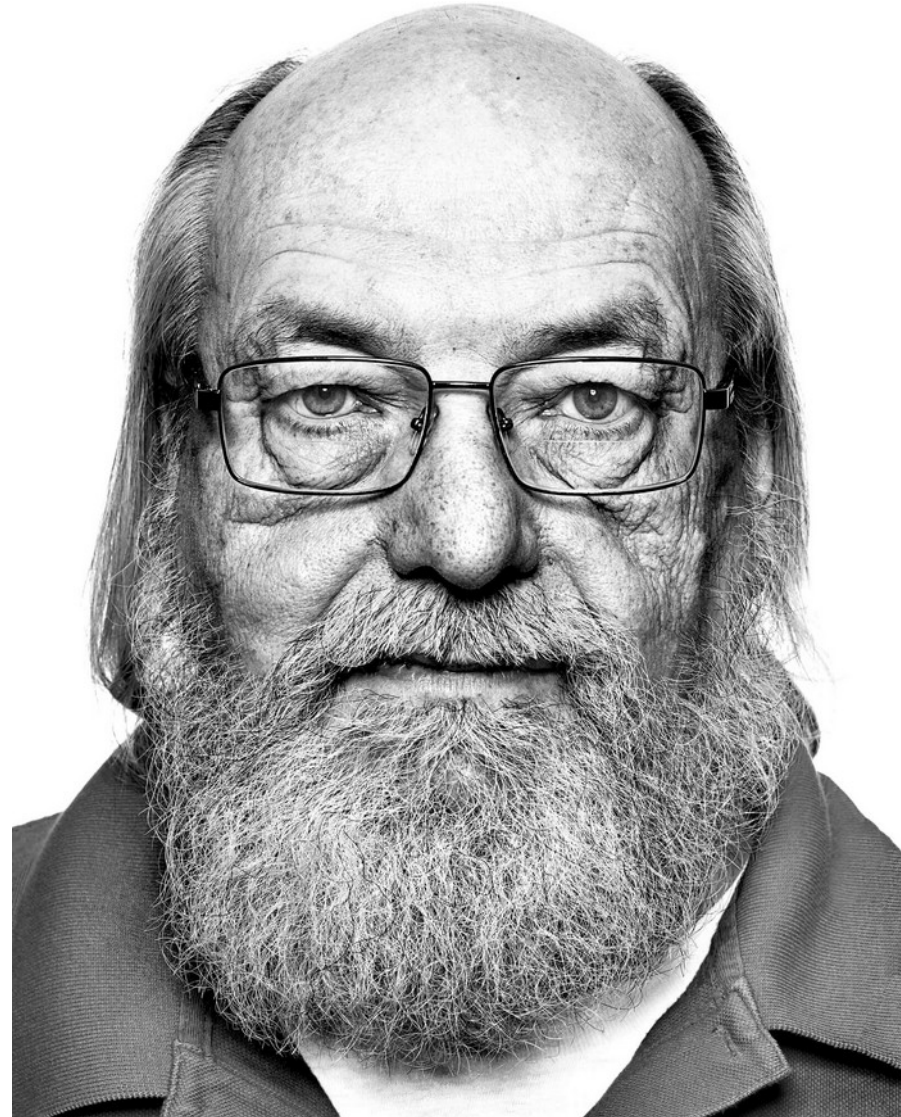


Multics

- MIT, General Electric, and Bell Labs (subsidiary of AT&T).
- Create a **multi-programming, multi-tasking, multi-user OS**.
- Developed between 1964-1975 (an early operational version was released in 1970).
- It, surprisingly, **failed**:
 - Expensive, arrive late to the market, limited availability, complex to use (only research facilities were interested in it).

Multics

- **Ken Thompson:** Concurrency (hashed semaphores, planning and process synchronization).



Multics

- Dennis Ritchie: System's programming language (he called it B).



Multics

- Thompson and Ritchie grew **disheartened with** the **Multics** Project.
 - Too complex and its development was messy.
- Thompson found a PDP-7 computer that was unused. He started to rewrite his game, **Space Travel**, to run in it.



Multics

- Thompson and Ritchie left the Multics project in 1969 and decided to start a hobby project within Bell Labs, based on Thompson's experience rewriting Space Travel:
- "Project Seven" aimed to develop an operative system with a clear philosophy in mind: **Portability (run anywhere) and simplicity (do only one thing, and do it well).**
- A new name for this project was later proposed by Brian KERNIGHAN, **UNIX**, mainly as **a pun on Multics.**

UNIX

- UNIX Version 1 was developed in 1971, using Assembly for the PDP-7 minicomputer.
 - It became very popular within Bell Labs.
- Thompson and Ritchie **received support from Bell Labs** to continue the development of UNIX.

UNIX

- Development of UNIX was **migrated** to a PDP-11/45.
- In **1972**, an improved and refined version of B was created as a system's programming language: It was called **the C programming language**.
 - Following the project's design philosophy, C focused on creating high level, portable, and efficient code.

UNIX

- In 1972, **UNIX Version 2** integrated the use of C as the system's programming language.
- In 1973, **UNIX Version 4** was **fully rewritten in C**.
 - Despite, Assembly will only be entirely removed from UNIX on Version 8 in 1985.
- However, things will get interesting with UNIX Version 5...

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THE UNIX WARD



The UNIX Wars

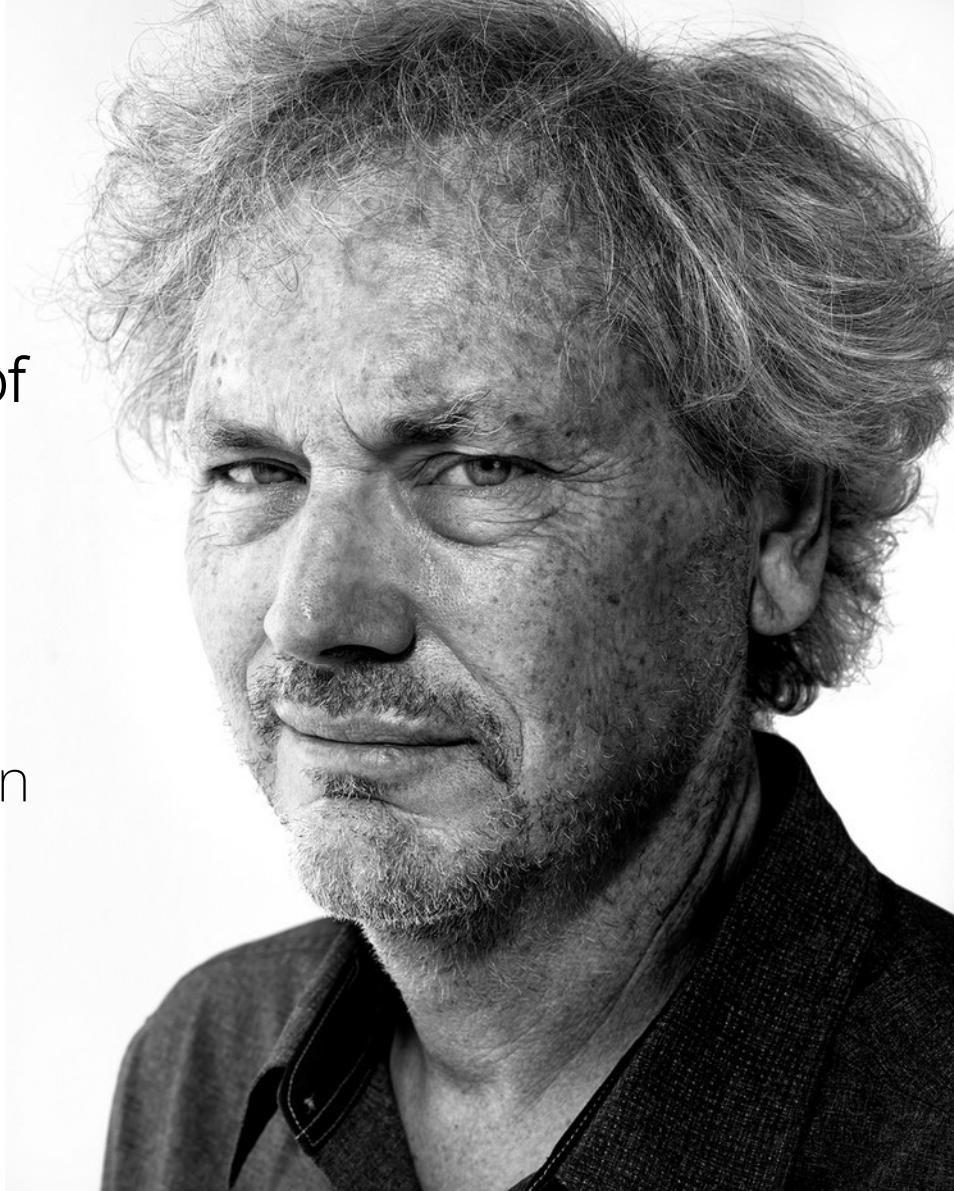
- AT&T VS U.S.: The anti-trust lawsuit.
 - AT&T was allowed to establish a **monopoly** on the U.S. **telephone** market.
 - In return, **AT&T inventions** will be **available** to academia **for free**, or by a fair and reasonable license.
 - AT&T will not be allowed to enter the computer business (hardware).

The UNIX Wars

- Unix Version 5 is the first **version that left Bell Labs**, and was distributed to selected U.S. institutions, and others that requested a copy.
 - No support for UNIX was provided by Bell Labs / AT&T.
 - This lead to University students and teachers to implement features and improvement for UNIX themselves.
- Thompson and Ritchie presented an article in an OS symposium, disclosing the development and functionality of UNIX.

The UNIX Wars

- Bob Fabry, professor at the **University of California Berkley**, Secured a copy of UNIX.
- **Bill Joy**, student at Berkley, finished the UNIX support for Pascal, that Thompson started.
 - He also implemented a text editor called ex, which will later become **Vi**.



The UNIX Wars

- This version of UNIX was labeled as the “Berkley Software Distribution UNIX” or **BSD**.
 - BSD grew vastly in popularity and copies of it were sold for \$50.
 - It became a **standard** for **academic institutions**.



The UNIX Wars

- In, **1979**, the Defense Advanced Research Projects Agency (**DARPA**) adopted BSD as the OS for their systems.
- Berkley consolidated their OS **dominance with 4.2 BSD**, the first OS to support ARPANET's protocol.
- In 1982, **Bill Joy** left Berkley and starts his own company called **Sun Microsystems**.
 - They created the **workstation computer**, featuring a custom OS called SunOS.
 - It was a computer capable of **networking**.

The UNIX Wars

- The popularity of UNIX exploded. **Companies** emerged **adapting UNIX** to a particular computer architectures.
 - This directly compromised UNIX's initial vision of portability.
- Even if BSD and Sun Microsystems were the responsible for popularizing and commercializing UNIX, **AT&T was not stale...**

The UNIX Wars

- Bell Labs never stopped the development and improvement of UNIX.
- By 1983, UNIX System V was commercially released as the latest and greatest version of UNIX.
- It was designed to become the standard version of UNIX.

The UNIX Wars

- AT&T took **drastic measures** to assume control of the UNIX market with System V:
 - System V integrated all additional **features** developed for 4.2 BSD.
 - A system of (very expensive) **licenses** were established **to limit the access** and development of UNIX by competitors.
 - Materials **documenting** the concepts and functioning of **UNIX** were **banned** (including even educational purposes).

The UNIX Wars

- These measures gave **a wound to UNIX**, from which it didn't recover:
 - The competition between **Berkley and AT&T** sparked a series of complex **lawsuits** between mid 1980s until mid 1990s.
 - The push for **System V and BSD** **divided the market** and made it unstable.
 - Due to the ever growing **variances of UNIX**, developing software for the platform became **unsustainable**.
- In **1987**, **AT&T and Sun Microsystems** **joined forces** in an attempt to unify the UNIX platform, merging their UNIX versions. **They failed**.

The UNIX Wars

- Despite these issues, there were **2 events** that finally “killed” UNIX...

The UNIX Wars

- The UNIX wars made UNIX and very unattractive platform, due to legal and financial instability.
- **Bill Gates** took advantage of this and positioned an OS of his own called Windows.





THE BIRTH OF LINUX

Linux

- Prof. Richard Stallman from M.I.T. decided to start the GNU project in 1983:
- Develop a truly **open-source** set of **packages** to (re)create an operative system (i.e. UNIX).
- Compatibility with the IEEE **POSIX**.



Linux

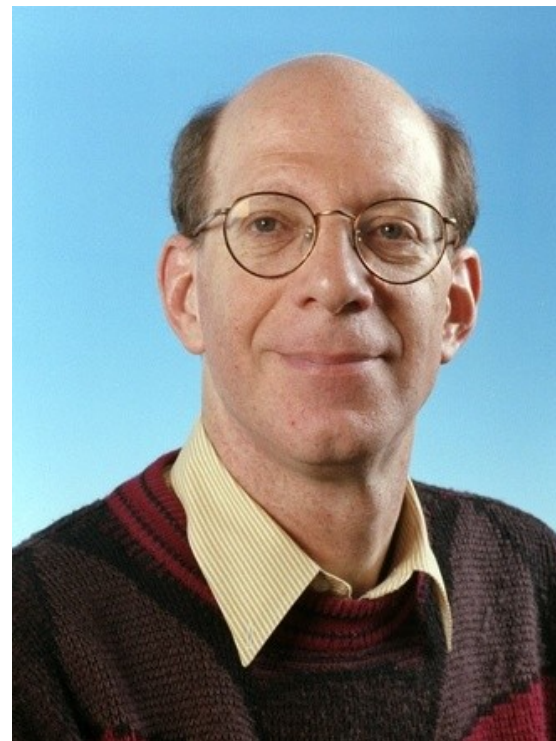
- In **1985**, Stallman creates the **Free Software Foundation (FSF)**, a non-profit to promote and protect free software.
- In **1989**, Stallman and FSF publish the **GNU General Public License**.
- By **1991** (8 year later), the GNU project had managed to create all necessary packages to create an OS, except one : **the Kernel**.
 - GNU HURD was the attempt at a kernel, but its development was stale.

Linux



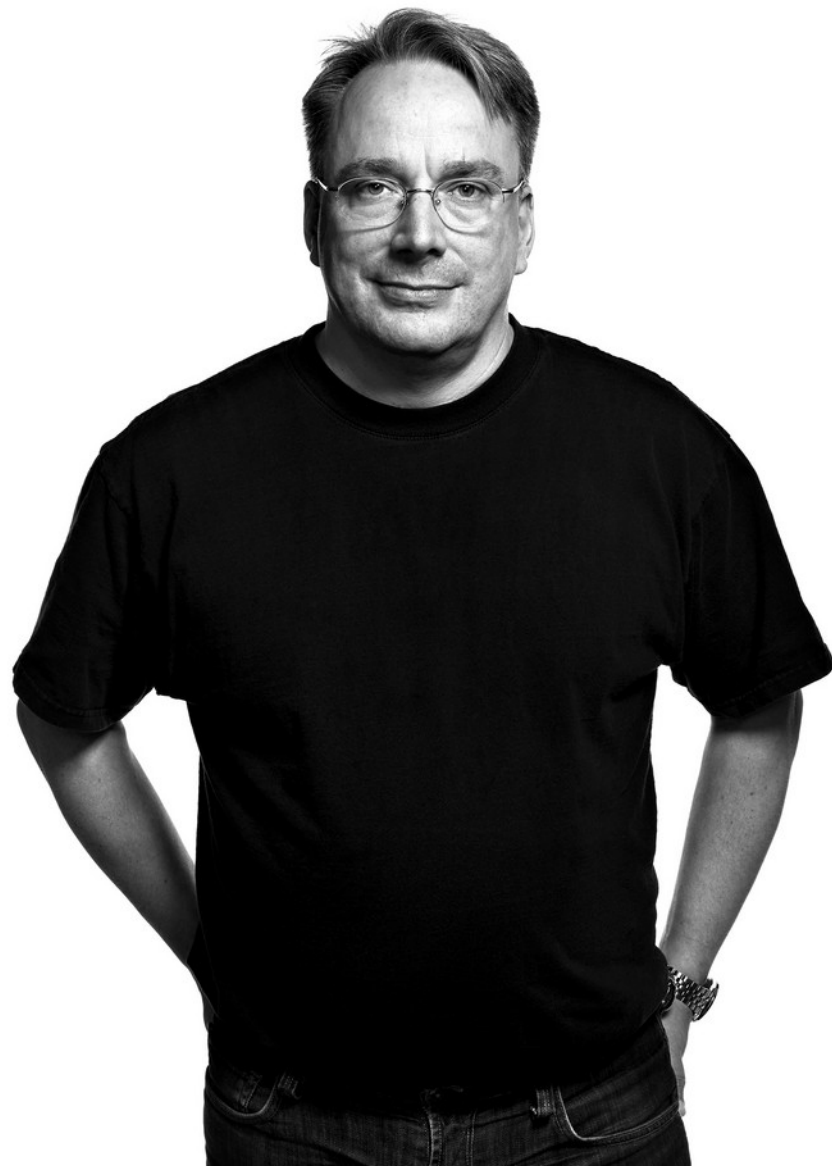
Linux

- University of Helsinki (1991).
- **Minix** by **Andrew Tanenbaum**.
 - OS for educational purposes.
 - Limited in capabilities, distribution, and modification.
 - Based on 16 bit architectures.
 - Intel's i386 processor (32 bit) was becoming the standard.



Linux

- Minix inspired **Linus Torvalds**, a CS student, to create a kernel that addressed its limitations:
 - Run on 32 bit architecture.
 - Free to distribute and modify.
- In August 25th, 1991, Linus **ask the Minix community** about desirable features for the kernel...



Linux

From: torvalds@klaava.Helsinki.FI (Linus Benedict Torvalds)
Newsgroups: comp.os.minix
Subject: What would you like to see most in minix?
Summary: small poll for my new operating system
Message-ID:
Date: 25 Aug 91 20:57:08 GMT
Organization: University of Helsinki

Hello everybody out there using minix -

I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu) for 386(486) AT clones. This has been brewing since april, and is starting to get ready. I'd like any feedback on things people like/dislike in minix, as my OS resembles it somewhat (same physical layout of the file-system (due to practical reasons) among other things).

I've currently ported bash(1.08) and gcc(1.40), and things seem to work. This implies that I'll get something practical within a few months, and I'd like to know what features most people would want. Any suggestions are welcome, but I won't promise I'll implement them :-)

Linus (torvalds@kruuna.helsinki.fi)

PS. Yes - it's free of any minix code, and it has a multi-threaded fs. It is NOT protable (uses 386 task switching etc), and it probably never will support anything other than AT-harddisks, as that's all I have :-).

Linux

- On September 21st, 1991, **Linus' 0.01 kernel** was released.
 - It was originally named Freax, but the name was modified by the FTP server admin (Ari Lemmke) to **Linux** (Linus + UNIX).
 - Linus invited anyone interested to contribute to the project.
- The project exploded in popularity world-wide!
- In 1992, the **Linux kernel** was released under the **GPL**.

Linux

- On March 14th, 1994, the **Linux 1.0.0 kernel** was released.
 - More than 12.000 people had contributed to the project at this point.
 - 176.239 lines of code.
- Providing most of the features offered by UNIX, but without any kind of license fees attach to it, **GNU/Linux overtook the UNIX customer base.**

Linux

- GNU and Linux started a **revolution** in the world of **computer science**:
 - The most used OS in the world.
 - The top 500 supercomputers run GNU/Linux.
 - 93% of the world's top web servers use GNU/Linux.

TO BE
CONTINUED

Take home message

- UNIX/Linux humble beginnings (in hardware and budget) lead to the portability, versatility, and efficiency that we experience in modern computing.
- To know UNIX and Linux is to know about the history and heritage of computer science.

Questions?

